

Notes: ClickHouse as a Real-Time Backend for Agent Memory — Relevance to SAW

Status: Design notes and open questions, based on external training attended Aug 18, 2026 (“The Agent’s Memory: Powering Real-Time AI Workflows with ClickHouse”). This is *not* an experiment report — no benchmarking has been run yet. It documents what was learned and how it maps onto the Sovereign AI Workforce (SAW) architecture, plus a concrete next step to actually test the idea.

Context

The Sovereign AI Workforce prototype currently designs its semantic-memory layer around PostgreSQL with the pgvector extension, coordinating seven LLM agents through a sequenced orchestration pipeline. This design has not been load-tested or benchmarked for retrieval latency at scale — a limitation already noted in the project’s evidence record.

The training session covered using ClickHouse (a column-oriented, real-time analytical database) as the data layer behind an AI agent, connected through a managed MCP server, with an emphasis on high-speed analytical queries feeding agent decisions.

What ClickHouse offers that pgvector-on-Postgres does not, in principle

- **Column-oriented storage**, optimized for fast aggregate/analytical queries across large volumes of structured event or log data — a different access pattern than the similarity search pgvector is built for.
- **Designed for real-time ingestion and query at scale**, which could matter if SAW’s agent memory grows to include high-frequency event logs (e.g., every tool call, every approval-gate decision) rather than just semantic embeddings.
- **MCP-server-based connection pattern**, which is architecturally similar to how SAW’s own agents would need to query a memory

backend — useful as a reference implementation pattern regardless of which database is chosen.

What this does NOT mean for SAW (explicit scope limit)

- This is not a claim that ClickHouse is “better” than pgvector for SAW’s use case. Semantic similarity search (the core job of pgvector in SAW) and high-speed analytical aggregation (ClickHouse’s strength) are different problems. They may be complementary rather than substitutes — e.g., pgvector for semantic retrieval, ClickHouse for logging/auditing agent decisions and approval-gate events at volume.
- No performance numbers exist yet. Any comparison beyond architectural reasoning would be speculation and is deliberately not included here.

Concrete next step (not yet done)

To move this from “notes” to “evidence,” the smallest honest next step is:

1. Take the existing SEAG experiment’s synthetic event log (already generated for the 1,700-trial experiment). 2. Load an equivalent copy into a local ClickHouse instance. 3. Run the same class of query SAW would need for an approval-gate audit trail (e.g., “all approval events for a given agent in the last N minutes”) against both the current PostgreSQL setup and ClickHouse. 4. Report actual latency numbers for both, including a negative result if ClickHouse shows no meaningful advantage at this data scale — consistent with how negative results are already reported elsewhere in this project (Axiom-Zero).

Until step 4 is done, this file remains a documented design consideration, not a validated architectural decision.